HW2

Monday, 17 February 2025 6:32 PM

Question 1

a) $\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = \nu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) + au^2$

→ Order = 2
→ Almost linear (au^2 adds non-linearity but all higher-order terms are linear)

b) $u_t + c\,u_x + a\,u^3 = \nu\, u_x\, u_{xx}$

→ Order = 2
→ Quasi-linear (highest order term is multiplied by a lower-order derivative)

c) $u_t + u_x^2 = \nu\, u_{xx}$

→ order = 2
→ Quasi-linear (u_{xx} does not have non-linear coefficients)

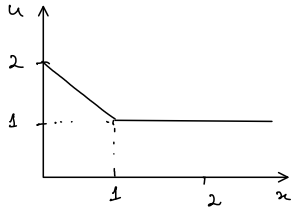
d) $u_t + u_x = \nu\,(u_{xx})^{0.5}$

→ order = 2
→ Non-linear

Question 2 & 4

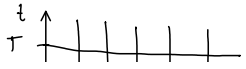
$u_t + u\,u_x = 0$ $\qquad u = \begin{cases} 2-x, & 0 \le x \le 1 \\ 1 & x > 1 \end{cases}$

$u(x=0) = 2$

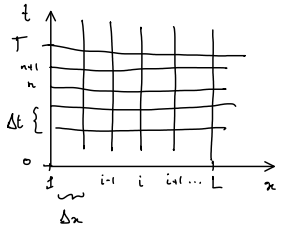


$\frac{u_i^{n+1} - u_i^n}{\Delta t} + u_i^n \frac{u_i^n - u_{i-1}^n}{\Delta x} = 0$; $\frac{2\Delta t}{\Delta x} \le 1$

→ Discretize the domain :



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→ Discretize the ODE :

time levels : $t^n = n\Delta t$; $n = 0, 1, 2 \ldots$

grid points : $x_i = (i-1)\Delta x$; $i = 1, 2, 3, \ldots m, m+1$

$$u(x_i, t^n)$$

Backward finite difference for u_x : $u_x = \dfrac{u_i^n - u_{i-1}^n}{\Delta x}$

Explicit Euler scheme for u_t : $u_t = \dfrac{u_i^{n+1} - u_i^n}{\Delta t}$

Rearranging the given equation yields :

$$u_i^{n+1} = u_i^n - \frac{\Delta t}{\Delta x}\left[u_i^n \left(u_i^n - u_{i-1}^n\right)\right]$$

Boundary conditions : $u(x=0, t) = 2$

Initial conditions : $u(x, t=0) = F(x) = \begin{cases} 2-x, & 0 \le x \le 1 \\ 1, & x > 1 \end{cases}$

$\Delta x = \dfrac{L}{nx}$; $nx = 100$; $L = 2$

$\Delta t = 0.001$; $nt = 100$ (number of time steps)

(a) Plot and evaluate solution

Algorithm

↳ First grid parameters are defined based on the stability constraints $\left(\dfrac{2\Delta t}{dx} \le 1\right)$

↳ Spatial grid of x w/ set dx points is initialized ;

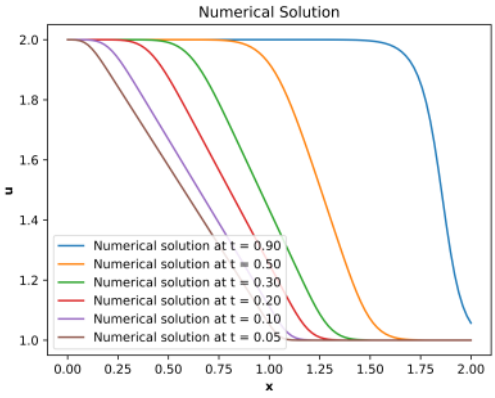
↳ Initial condition for u is established w/ the given function

↳ Nested for loop where outside loop steps the time while inside loop calculates the (n+1) value :

$$u_i^{n+1} = u_i^n - u_i^n \cdot \frac{\Delta t}{\Delta x}\left(u_i^n - u_{i-1}^n\right)$$

↳ The boundary condition is applied, where $u(x=0, t) = 2$

↳ results are plotted for varying time steps ; where $0 < t < 1$



Nonconservative method (w/ the upwind scheme) will not converge to a discont. weak solution as the grid is refined

(b) Plot exact solution via characteristics curves in the x-t plane

Characteristic curve solution → $\frac{dx}{dt} = u(x,t)$; $\underbrace{\frac{du}{dt} = 0}$

u is a constant along the characteristic equation

$x = ut + C_1$

$C_1 = x - ut$

$u = C_2$

Characteristics are straight lines ; C_1 must be a function of u ;

Hence $u = C_2(x - ut)$

$u = F(x - ut)$, where $F(\xi)$ is the provided initial condition @ $t=0$

$x = F(\xi)t + \xi$, where $\xi = \xi(x,t)$ & is the x-intercept of characteristic curve

$0 \leq x \leq 1$; $F(\xi) = 2 - \xi$

$t = \frac{x - \xi}{F(\xi)}$

$x > 1$; $F(\xi) = 1$

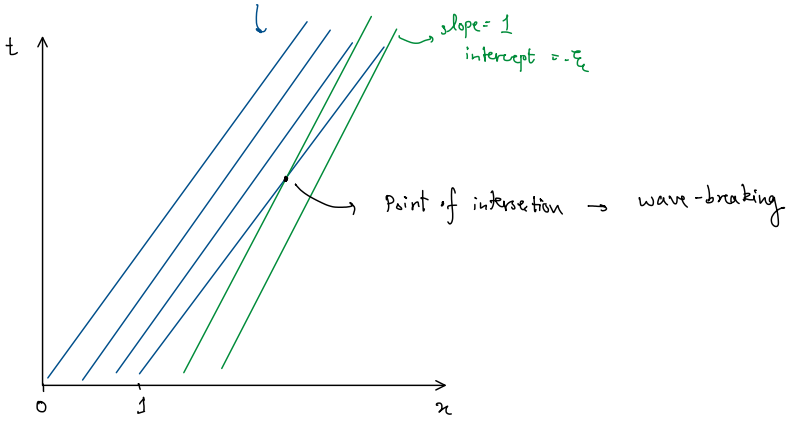
$t = x - \xi$

$t = \frac{x - \xi}{2 - \xi}$

$0 \le \xi < 2 \rightarrow$ positive slope w/ negative t-intercept

$\xi < 0 \rightarrow$ positive slope (less steep) w/ positive intercept

slope $\left(\frac{1}{2-\xi}\right) \rightarrow$



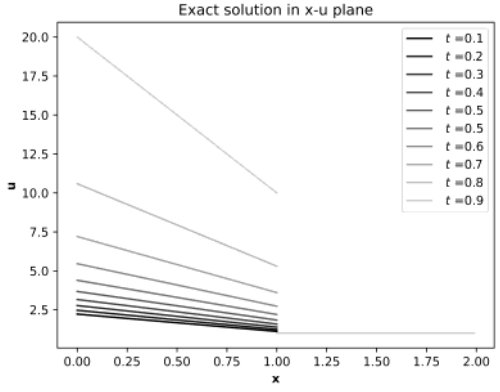
& $u(x,t) = F(x-ut)$

given, $u(x,0) = F(x) = \alpha x + \beta$

$u(x,t) = \alpha(x-ut) + \beta$

$$\boxed{u(x,t) = \frac{\alpha x + \beta}{\alpha t + 1}}$$

For each fixed time, the solution is a straight line w/ slope $\left(\frac{\alpha}{\alpha t+1}\right)$ in the x-u plane



The above graph shows the solution for u where for region

$0 \le x \le 1$; $u(x,t) = F(x-ut) = \frac{2-x}{1-t}$

As t approaches 1, the solution blows up and for varying t the the slope of line is changing i.e. as t increases the slope (and intercept) increases (since $(1-t)$ gets smaller so $\frac{1}{(1-t)}$ gets bigger)

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$x > 1 \quad ; \quad u(x,t) = 1$

This is just a constant u line for all values of x & t

Therefore $x=1$ forms the discontinuity in the solution

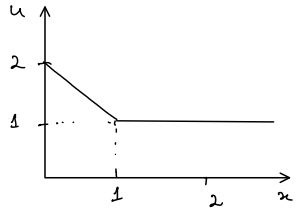
(c) Explain where BC are needed by exact & by numerical solution

For numerical solution, the initial nodes are $u_i^0 = u_0(x_i)$ which is the given expression in the problem. The boundary node is the given $u_0^n = u_1(t^n)$

Question 3 & 4

$$u_t + \frac{\partial(u^2/2)}{\partial x} = 0 \qquad u = \begin{cases} 2-x, & 0 \le x \le 1 \\ 1 & x > 1 \end{cases}$$

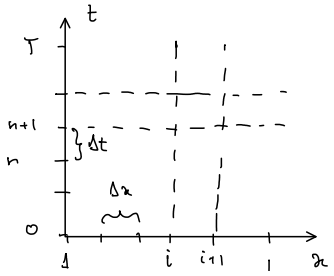
$u(x=0) = 2$

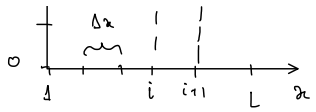


$$\frac{u_i^{n+1} - u_i^n}{\Delta t} + \frac{(u_i^n)^2 - (u_{i-1}^n)^2}{2\Delta x} = 0 \qquad ; \quad \frac{2\Delta t}{\Delta x} \le 1$$

(a) Plot and evaluate solution

• Discretize the domain :





- Discretize the ODE:

time levels : $t^n = n\Delta t$; $n = 0, 1, 2 \cdots$

grid points : $x_i = (i-1)\Delta x$; $i = 1, 2, 3 \cdots m, m+1$

$$u_i^{n+1} = u_i^n - \left[\frac{(u_i^n)^2 - (u_{i-1}^n)^2}{2\Delta x} \right]$$

Conservative PDE of the form $u_t + [f(u)]_x = 0$

can be discretized as such:

$$u_i^{n+1} = u_i^n - \frac{\Delta t}{\Delta x} \left(f(u_i^n) - f(u_{i-1}^n) \right)$$

$nx = 100$ $\quad L = 2$

$dt = 0.001$

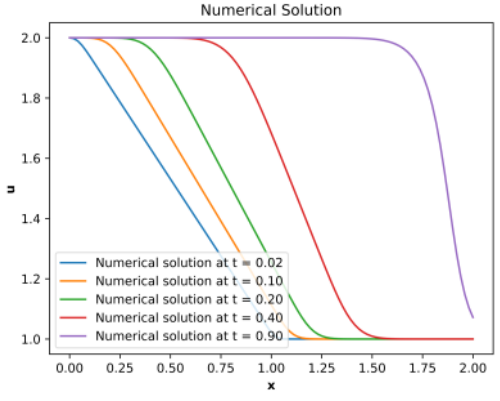
Algorithm:

- A similar algorithm to Q3 is followed w/ some key changes to include the upwind conservative algorithm mentioned above

$$f_plus = \frac{1}{2}(u_i^n)^2$$

$$f_minus = \frac{1}{2}(u_{i-1}^n)^2$$

$$u_i^{n+1} = u_i^n - \frac{\Delta t}{\Delta x}(f_plus - f_minus)$$



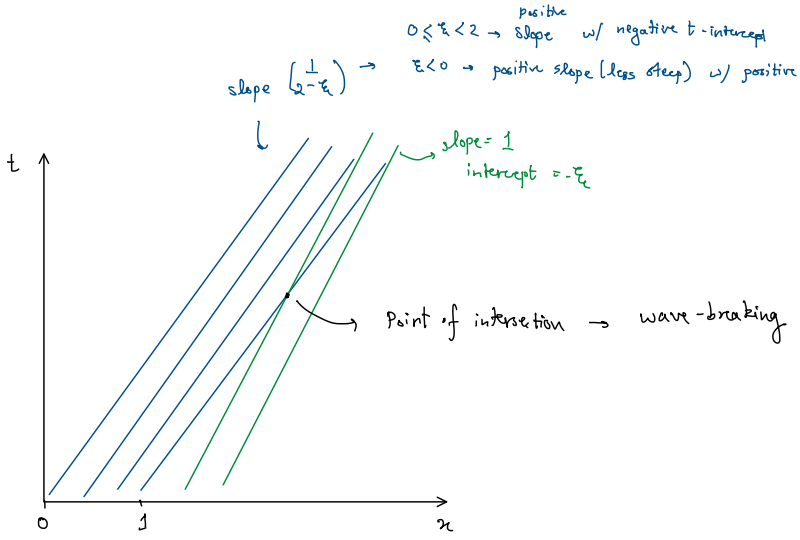
(b) Plot exact solution via characteristics curves in the x-t plane

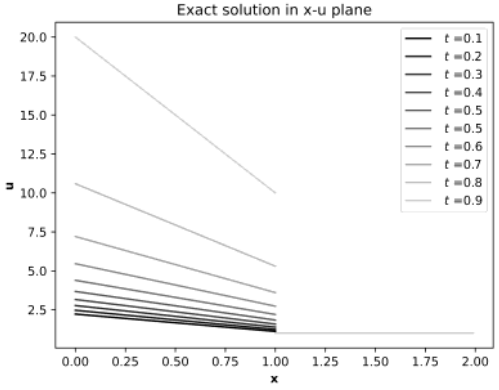
parameterizing by s:

$x(s,0) = s$; $\quad u(s,0) = u_0(s)$

$\frac{\partial x}{\partial t} = u$, $\frac{\partial u}{\partial t} = 0$

$x = u_0(s)\,t + s$ → will be similar to the previous analysis for non-conservative equation. Exact solution is given by the characteristic curves ($x-t$ & $x-u$)





(c) Explain where BC are needed by exact & by numerical solution

The numerical solution uses the same boundary & initial condition nodes: $u_0^n = u_1(t^n)$ & $u_i^0 = u_0(x_i)$

The exact solution gives infinite no. of curves, using the given initial conditions, where a specific solution can be found by using the BC.

Question 4 (contd.)

(a), (b), (c) → see previous parts

(d) Explain weak solution concept. Contrast shock speed

Shocks are a result of intersection of characteristic curves of a given PDE. If the curves are paralell then solution exists for all t. A breaking time of less than 0 can result when the characteristic lines fan out; this results in a global solution for t>0 and a phenomena called rarefaction wave. For initial condition where the condition defined in left domain is larger than than that in the right hand side domain then this would lead to a shock wave where shock speed is the speed of propagation of this wave.

Smooth solutions of conservation laws can break down in a finite amount of time due to intersections in x-t plane (as shown previously). Once a shock occurs, classical solutions are no longer applicable, necessitating a new approach which is termed as weak solution. The involves defining a solution after the breaking point by allowing for u to be discontinuous.

For a weak solution, it must still satisfy some criteria's (i.e. Rankine-Hugonoit jump condition) across the discontinuity curve. This condition gives the shock speed $s = \frac{f(u_L)-f(u_R)}{(u_L-u_R)}$ which is equivalent of saying the ratio of "jump in f(u)" over "jump in u" (from left to right).

(e) Explain entropy condition

There may be cases when the weak solution does not give unique solution after the shock. This is where an entropy condition can be used to disregard non-physical solutions.

The condition allows for a discontinuity only when the left wave is moving faster than the right wave. Therefore, if a wave is propagating with speed s where $s = \frac{dx}{dt}$ then $u^+ < u^-$. This is mathematical equivalent of saying that every physical process the entropy of the system should not decreasing (which is directly related to the 2nd law of thermodynamics).

Question 5 Eigenvalue approach

(a) $5u_{tt} - 7u_{xx} + 8u_{yy} + u_{zz} = 0$

$$A = \begin{bmatrix} 5 & & & \\ & -7 & & \\ & & 8 & \\ & & & 1 \end{bmatrix}$$

Eigenvalue of diagonal matrix are just the diagonals

∴ Hyperbolic
($\lambda_i \neq 0$ & all but one have same sign)

(b) $5u_{tt} + 7u_{xx} + 8u_{yy} + u_{zz} = 0$

$$A = \begin{bmatrix} 5 & & & \\ & 7 & & \\ & & 8 & \\ & & & 1 \end{bmatrix} \qquad \lambda_i = \{5, 7, 8, 1\}$$

∴ Elliptic
(λ_i all have the same sign)

Question 6 Canonical-form approach ; independent vars $\{x, y, z, t\}$

(a) $5u_{tt} - 7u_{xx} + 8u_{yy} + u_{zz} = 0$

Highest order derivative exists for all w/ one different sign → Hyperbolic

(b) $5u_{tt} + 7u_{xx} + 8u_{yy} + u_{zz} = 0$

Highest order derivatives of all vars but all are positive → elliptic

(c) $5u_t - 7u_{xx} + 8u_{yy} + u_{zz} = 0$

highest order derivatives for $\{x, y, z\}$ but t

parabolic in "broad sense"

(d) $xu_{tt} + 7u_{xx} + 8u_{yy} + u_{zz} = 0$

Mixed, need to evaluate determinant of coefficient matrix where the sign of x will determine the property

Question 7

↳ BC's & IC's by 2nd order elliptical, parabolic & hyperbolic PDE's

↳ weak solutions;

Hyperbolic 2nd order PDEs:
Characteristic curves are real and distinct where theres exists domains of influence and dependence for a solution at each node in the domain. Therefore, initial and boundary conditions can be imposed only on where the characteristic base curves enter the domain. The number of conditions is equal to the number of characteristics that can enter the domain at that point. All characteristics passing through the surface need to have their IC/BC defined in order to completely specify conditions for a solution

They are capable of having weak solutions and the nature of these is similar to that of the first order PDE's

Parabolic 2nd order PDEs:
An initial value for the solution must be given at every point of the spatial domain at t=0 and the boundary value for the solution must be given at every point on the boundary of the spatial domain for every instant of time $0 \leq t \leq t_0$ to determine solution up to time t_0

Parabolic do not have weak solutions and any discontinuities are eventually removed so that the solution and its derivatives are all continuous in the domain interior.

Elliptical 2nd order PDEs:
Boundary condition is needed at every point on the domain as these PDEs do not have real characteristics. However, initial conditions are irrelevant since there are no time-like coordinates. These PDEs do not have weak solutions and discontinuities in BC are not propagated into the domain.

Above answers have been paraphrased from Shih-Caughey, Chapter 4